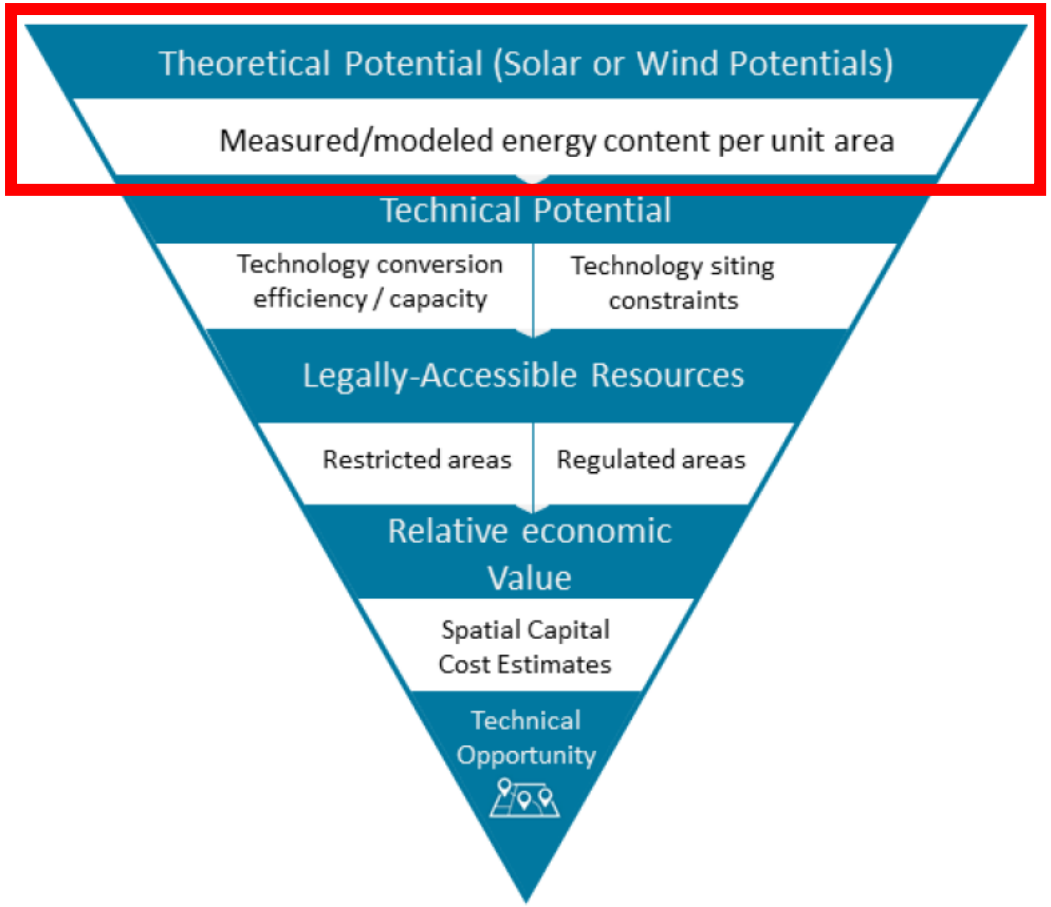
GIS Site Analysis for a Hypothetical Solar Energy Farm

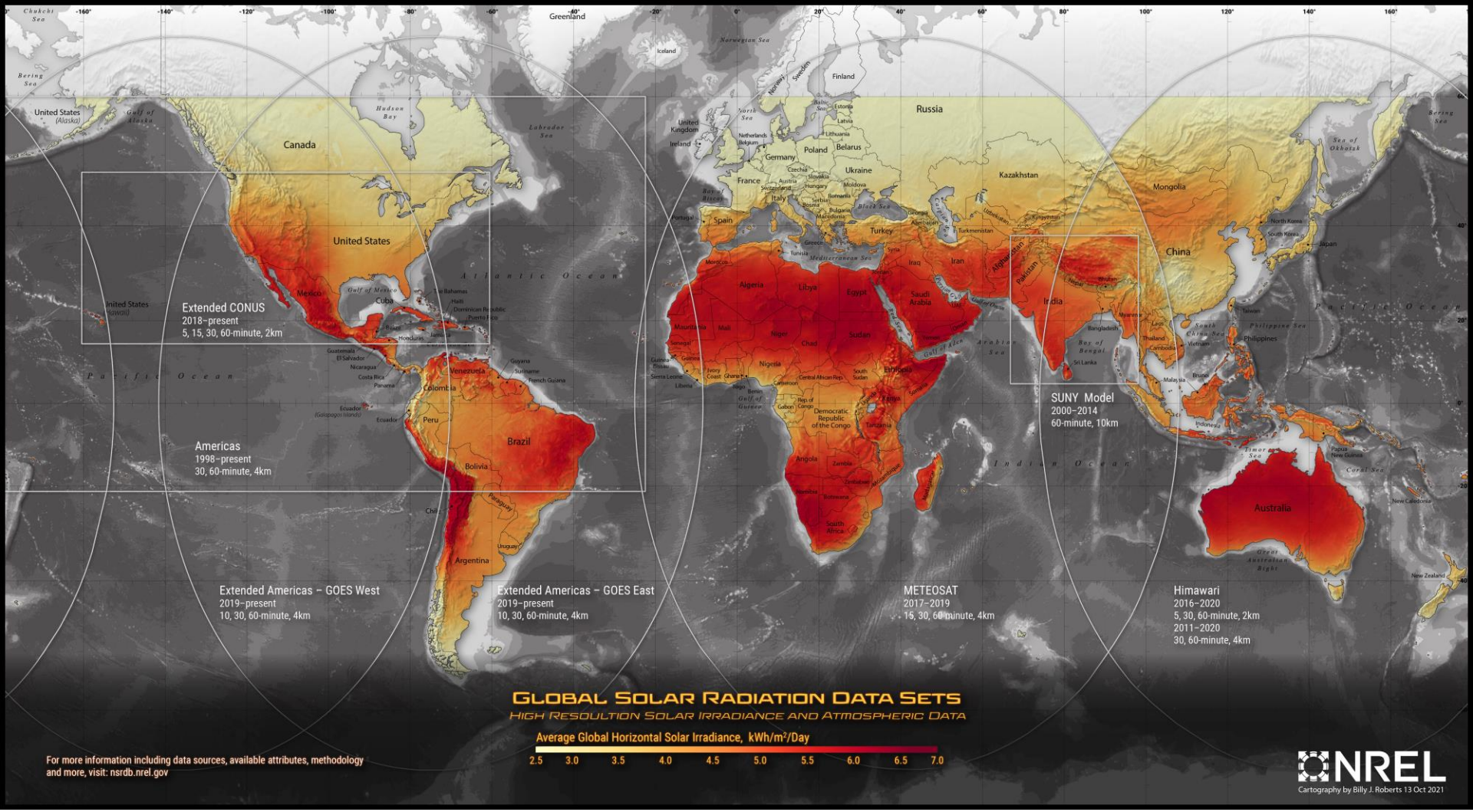
- **Theoretical Potential:** amount of solar energy
- **Technical Potential:** technology, and site characteristics
- **Legally Accessible:** Alberta's wildlife habitat sensitivity risk
- **Relative Economic Value:** costs to access/prepare site

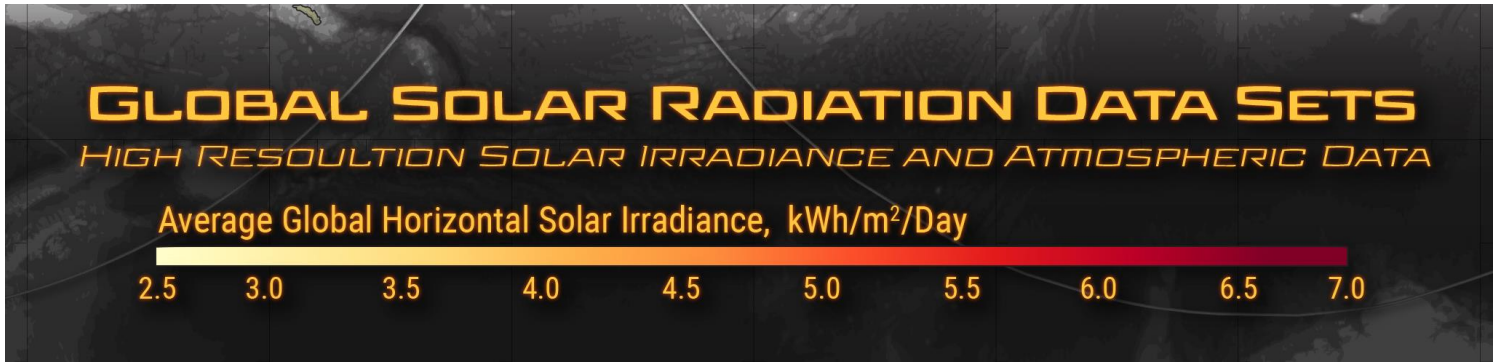
Theoretical Potential



"Measured or modeled energy potential across a geographic area."

How much solar radiation is received at any given location? Depends on latitude (day length, seasons), and cloud cover.


Extended CONUS
2018–present
5, 15, 30, 60-minute, 2km
Americas
1998–present
30, 60-minute, 4km
Extended Americas – GOES West
2019–present
10, 30, 60-minute, 4km
Extended Americas – GOES East
2019–present
10, 30, 60-minute, 4km
METEOSAT
2017–2019
15, 30, 60-minute, 4km
SUNY Model
2000–2014
60-minute, 10km
Himawari
2016–2020
5, 30, 60-minute, 2km
2011–2020
30, 60-minute, 4km
GLOBAL SOLAR RADIATION DATA SETS
HIGH RESOULTION SOLAR IRRADIANCE AND ATMOSPHERIC DATA
Average Global Horizontal Solar Irradiance, kWh/m²/Day
2.5 3.0 3.5 4.0 4.5 5.0 5.5 6.0 6.5 7.0
For more information including data sources, available attributes, methodology and more, visit: nsrdb.nrel.gov
NREL
Cartography by Billy J. Roberts 13 Oct 2021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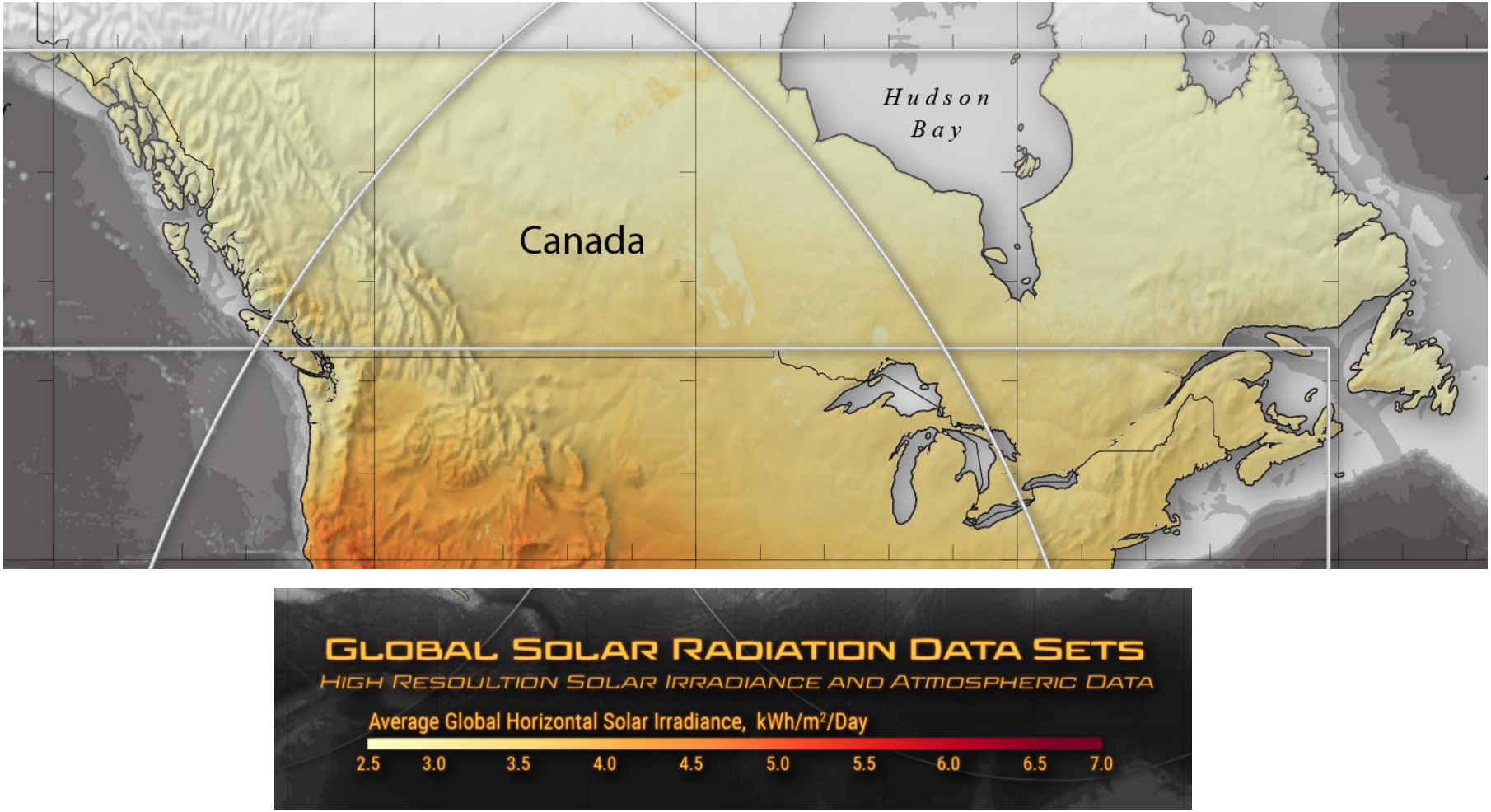
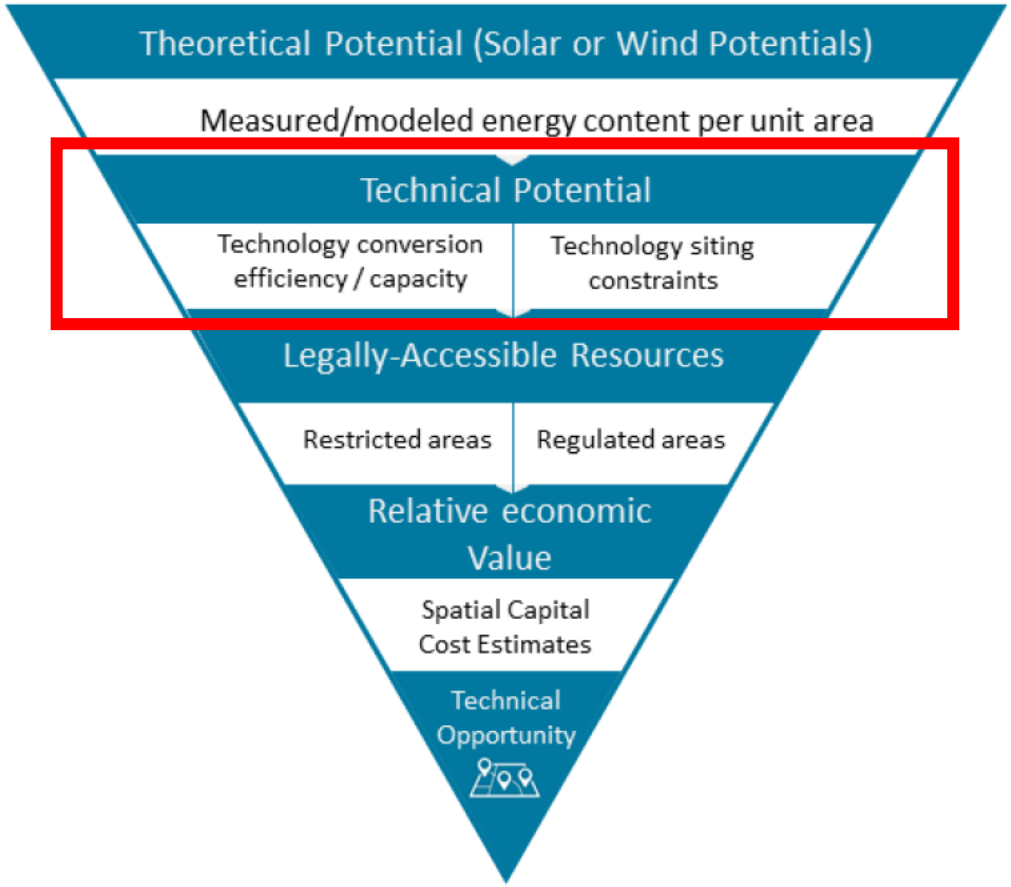


For a small area within Alberta, the theoretical potential solar energy resource would be essentially equal across the area. Such global radiation data would be useful to compare countries across the globe.

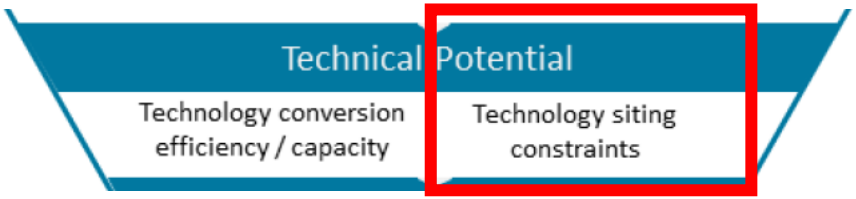
Technical Potential



Those "theoretical resources" that can be converted into useful energy by prevailing technologies (**technology conversion efficiency** limit), at sites that can be accessed using reasonable engineering solutions (**technology siting constraints**)."

Both are dependent on interactions between the technology and site characteristics.

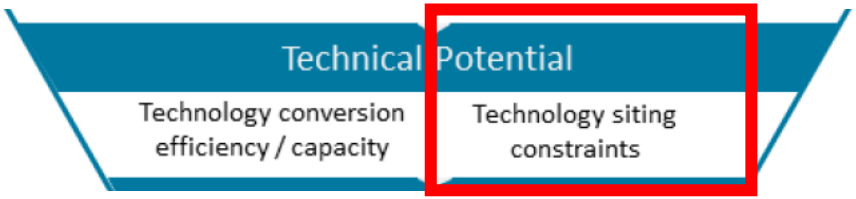
Technical Potential Cont'd



Technical siting constraints consider those areas (of land) that are technically feasible to reach/access and receive solar radiation in quantities viable for current technology.

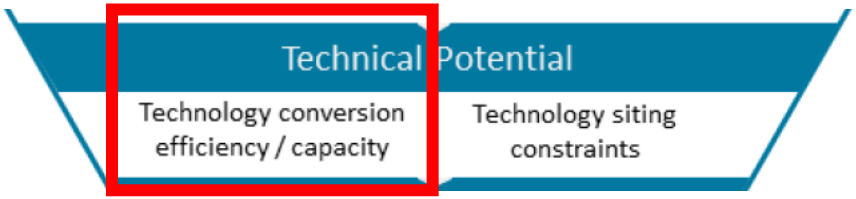
- **Land and not water** (not looking for fluviovoltaic locations)
  - work within the Nation's boundary (outline)
  - Remove water bodies and streams (and area near water) from consideration
- **Slope:** not too steep – installation/access
  - Digital Elevation Model (DEM): process
  - derive slope and choose areas with appropriate slope values

Technical Potential Cont'd



- **Aspect:** any southern exposure for Northern Hemisphere location (not north facing)
  - derive aspect from the DEM and choose areas with appropriate aspect values
- **Land Cover:** Already developed? Forested? Crop land? Barren wasteland?
  - There may be a desire to save good soil for agriculture, or protect existing forested land.
  - Different land covers have different costs associated with clearing/preparing the land for development
- **(Assignments do not include aspect or land cover)**

Technical Potential Cont'd



The **technology conversion efficiency refers to the** "capacity factor" and conversion efficiency.

Total capacity (MW) = $\frac{\text{Total Area (ha)}}{\text{LUI (ha/MW)}}$

- Land Use Intensity (LUI) typically ranges from 2 to 5.
- Research in Canmore used a LUI of 4.

Watt: unit of power.
Megawatt: 1 million Watts.
Land-use intensity: the area of land required per unit of energy production.

Technical Potential Cont'd

Total Generation =	Total Capacity (MW)	x	Hrs/Yr (8760) (hrs/yr)	x	Capacity Factor (%)

Capacity Factor:

- Usually ranges between 10 to 25%.
- Research completed in Alberta* used a capacity factor of 22%.

* Miller, L.M.; Keith, D.W. Corrigendum: Observation-Based Solar and Wind Power Capacity Factors and Power Densities (Environmental Research Letters (2018) 13 (104008). Environ. Res. Lett. 2019, 14.

Technical Potential Example

Total capacity = $\frac{\text{Total Area (ha)}}{\text{LUI (ha/MW)}} = \frac{1000\text{ ha}}{3.5\text{ ha/MW}} = 285.7\text{ MW}$

(Land area of 1000 ha for this example.

Using a LUI of 3.5; half-way between 2 and 5.)

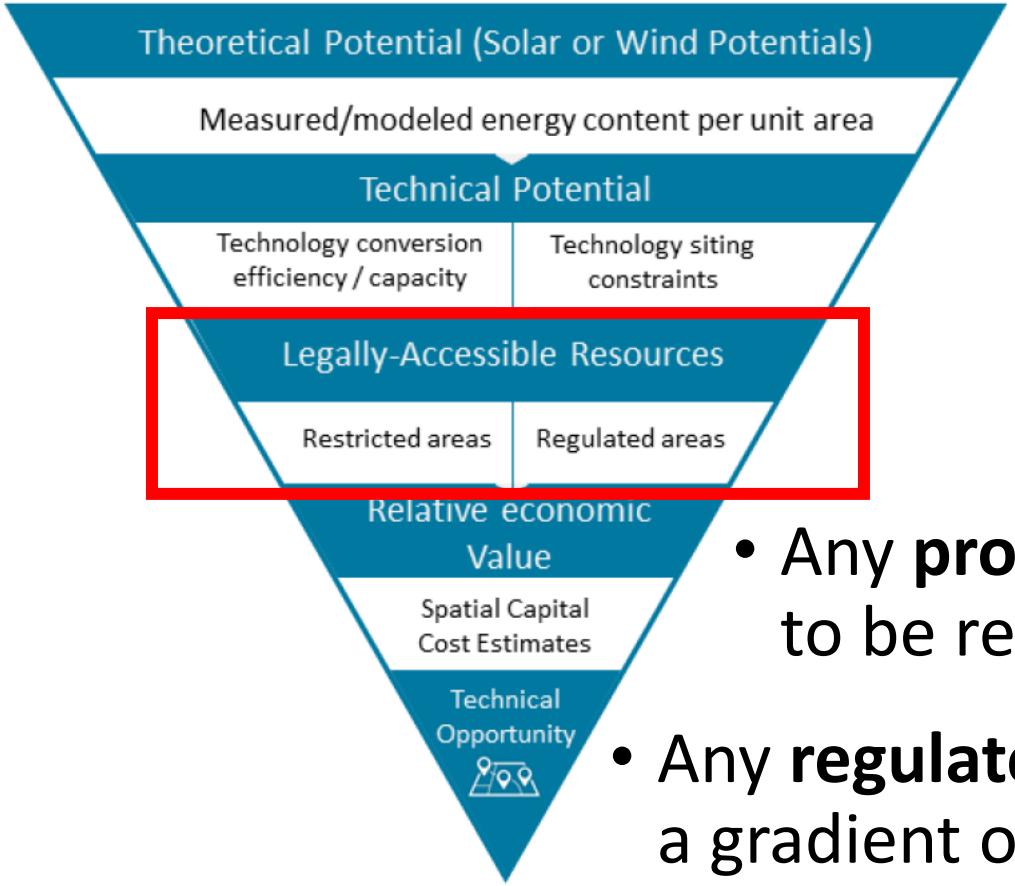
Total Generation = Total Capacity (MW) x Hrs/Yr (8760) (hrs/yr) x Capacity Factor (%)

= 285.7 MW x 8760 hrs/yr x 0.22

= 550,601 MWh/yr

(Using a capacity factor of 22% based on Alberta research.)

Legally Accessible Resources



Land areas might be 'technically feasible', but existing regulations regarding land-use and infrastructure siting cannot be violated.

- Any **prohibited areas** would need to be removed from consideration.
- Any **regulated areas** are mapped along a gradient of 'regulatory risk' from high to low, based on the likelihood of government to approve or decline a proposal.

Legally Accessible Cont'd

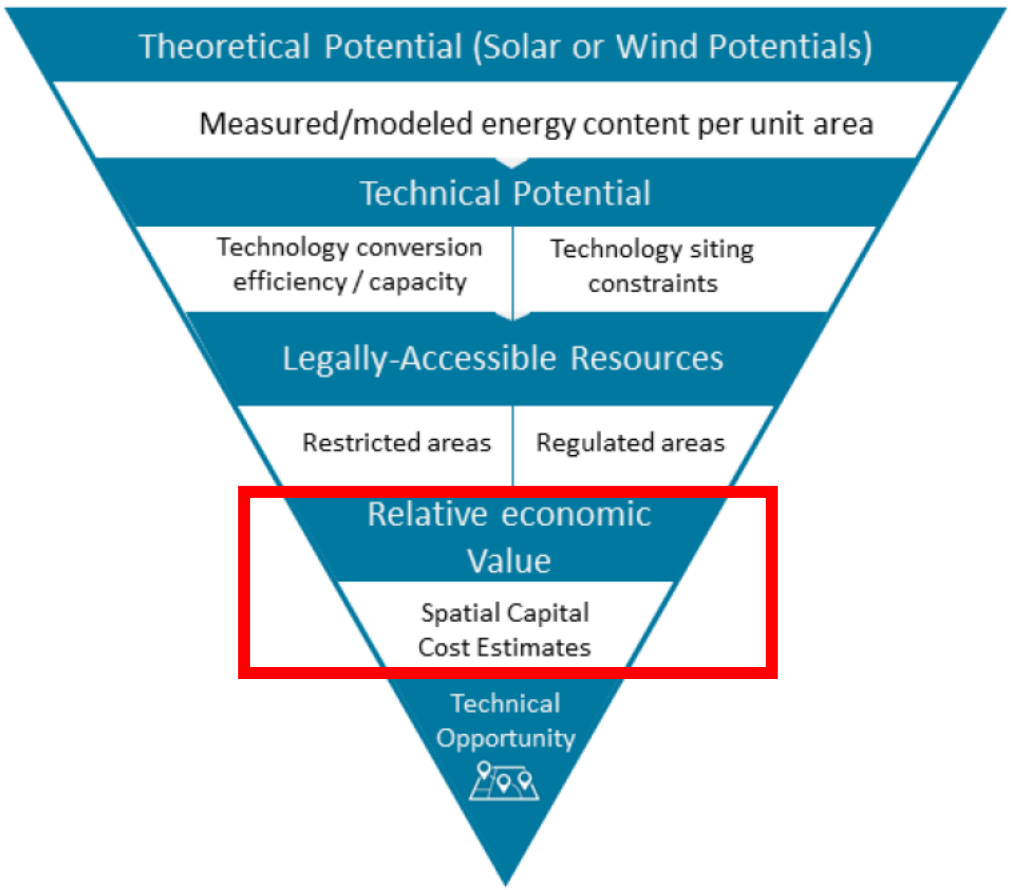
- **Restricted:** can mean total blockage
- **Regulated:** means controlled
- **Cultural and natural heritage sites**, **parks** and otherwise **protected areas** would not be considered legally accessible. (restricted?)
- **Zoning bylaws** might prevent certain types of development, including some forms of renewable energy. (restricted?)
- **Sensitive wildlife habitat** must be avoided, to reduce wildlife mortality, habitat destruction and future mitigation and related costs.

Legally Accessible Cont'd

Wildlife Habitat Sensitivity Risk:

- Alberta has habitat categories, ranked (ie. an ordinal scale):
    - Critical,
    - High,
    - Moderate, and
    - Lower risk categories
- proposed development sites are more likely to succeed if planned for low or moderate risk areas.
- Critical wildlife zones will be removed from consideration

Relative Economic Value



Spatial **capital cost estimates** provide the information required to determine the **economic viability** of a site.

Road access, the transmission and distribution of electricity, and land cover are considered here.

Relative Economic Value Cont'd

Roads:

- The closer a solar farm is to existing roads, the more economical it will be.

Transmission Lines:

- The closer a solar farm is to existing electricity infrastructure, the more economical it will be.

Both roads and transmission lines have costs associated with their extension into suitable land areas.

(Once again, land cover clearing costs are not considered in the assignments.)

GIS Analysis - 1

Indian Reserve

- The area (or outline) that identifies where a solar farm is being considered is created.

Roads, Streams and Lakes

- These features and the area ‘near’ these features are identified (for later removal from consideration).

Wildlife Habitat Sensitivity Risk

- WHSR data is explored.

(PreProcessing)

- All new data is preprocessed for correct spatial coordinates, and reduced to the area of the Indian Reserve.

GIS Analysis - 2

DEM

- The DEM is **preprocessed** for correct spatial coordinates.
- The DEM is explored for its **raster properties** (before and after preprocessing).
- **Hillshade** and **slope** terrain products are created, as well as contour lines.
- **Steep slopes** are isolated (for later removal). This involved vectorization of the raster data.

GIS Analysis - 3

Transmission Lines and Associated Costs:

- An image of transmission lines is **georeferenced** (ie. made to have spatial coordinates).
- Transmission line data in the form of a line shapefile is created through **heads-up digitizing**.
- A **transmission line cost distance raster** is created, ie. the costs of extending tr. lines from existing network.

Road Costs:

- A **road cost distance raster** is created, indicating the costs of extending roads from existing road network.

Road and transmission line cost distance rasters are summed together.

GIS Analysis - 4

Wildlife Habitat Sensitivity Risk:

- Low and Moderate WHSR areas are kept for consideration (ie. **legally feasible land**)

Road/TrLine Costs:

- Costs are subdivided into Low/Mod/High cost classes,
- Low and Moderate Cost areas are kept for consideration (ie. **economically viable land**)

Legally feasible and economically viable land:

- Must find land that is BOTH legally feasible AND economically viable.
- Represents the ‘possible’ land (so far); **Legal/Econ**

GIS Analysis – 4 Cont'd

Roads, Streams and Lakes:

- The areas of roads, streams and lakes, and their 'near' areas are created and amalgamated together with the land that is too steep for development.
- This represents the **non-technically feasible land**.

- Non-technically feasible land is removed from the possible land (Legal/Econ), resulting in the land that can **potentially be developed**.

GIS Analysis – 5

Potential Sites for Further Exploration:

- Heads-up digitizing of sites, based on the land that can **potentially be developed**.
- Also consider the **current land use** (homes, buildings); **imagery** basemap of ArcPro used to avoid these sites.

- Determine/**create attributes** for these potential sites:
  - **Area, total capacity, total generation**
  - **Road/TrLine cost** to access parcel (involving creation of SAT and joining to shapefile)
  - **Cost per total generation**